Sales Trend Analysis and Revenue Forecasting System

Using Python & Power BI

Introduction

In today’s competitive retail environment, understanding sales trends and predicting future revenue is crucial for business growth. This project focuses on analyzing historical sales data and forecasting future revenue using data analytics and visualization tools.

Why I Chose This Project

- Retail industry heavily depends on sales forecasting
- Businesses need data-driven decision making
- I wanted to apply Python + Power BI together
- Real-world application of Data Analysis
- Combines Database + Visualization + Forecasting

Problem Statement

Retail companies generate large amounts of sales data but struggle to:

- Identify trends
- Detect seasonal patterns
- Understand external factors impact
- Predict future revenue

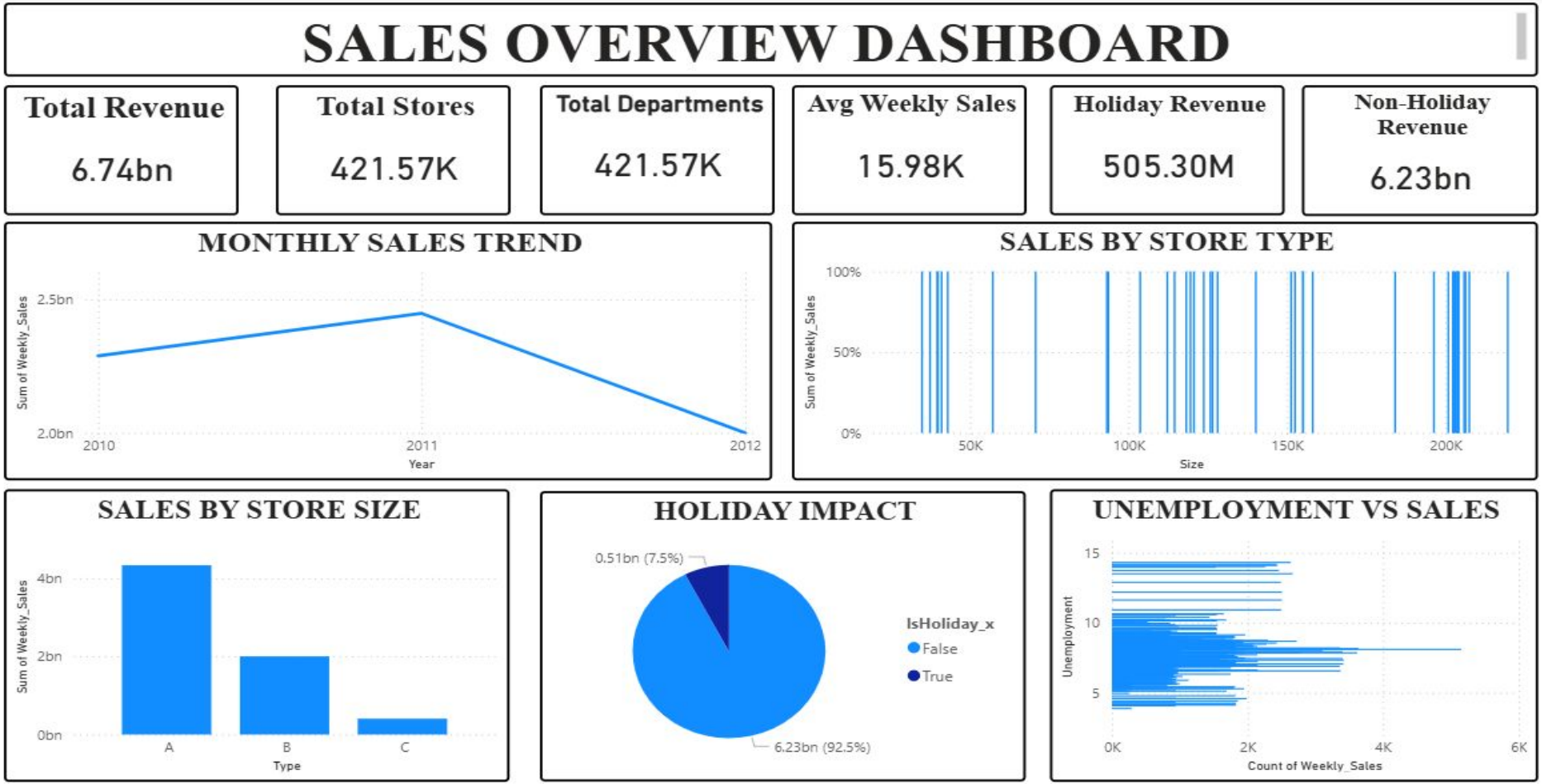
Objectives

- Perform Exploratory Data Analysis (EDA)
- Clean and preprocess raw sales data
- Analyze sales trends
- Identify external factor impact
- Build a forecasting model
- Design interactive Power BI dashboard

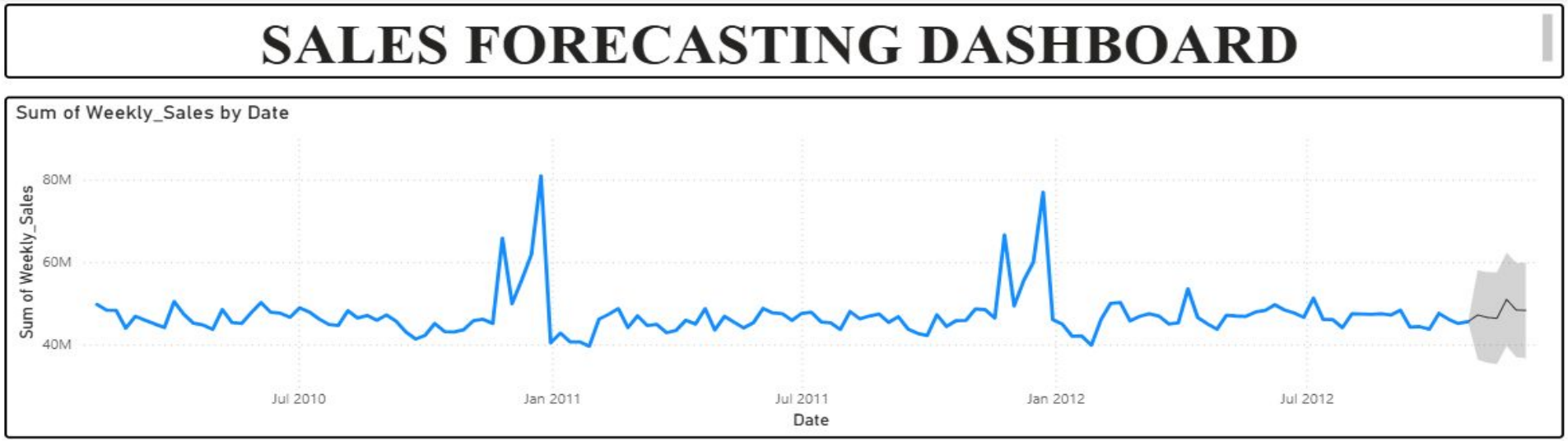
Dataset Overview

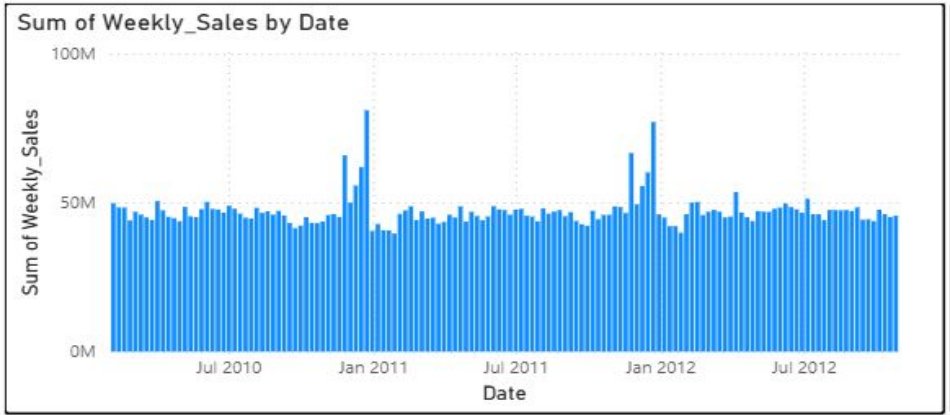
- 421,000+ records
- Store-level sales
- Department-wise sales
- Date-wise sales
- Holiday indicator
- External economic factors

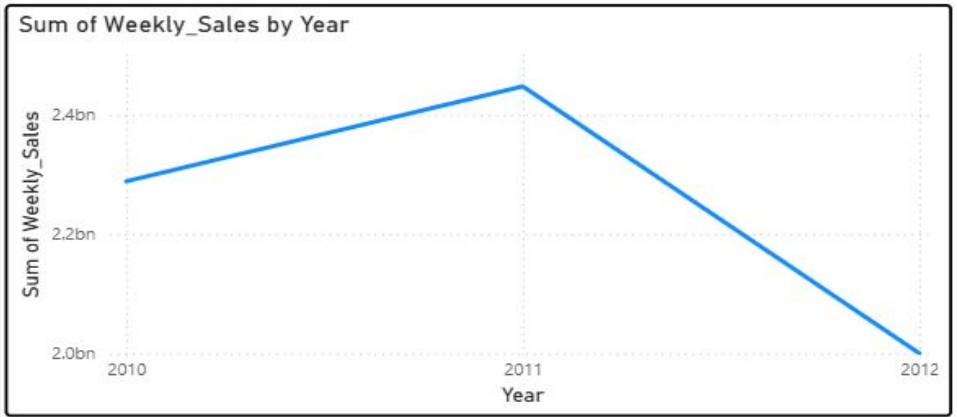
Power BI Dashboard



Power BI Dashboard







Power BI Dashboard

Key Insights

- Holiday season increases revenue
- Large stores generate higher sales
- Forecast shows stable growth trend
- Economic factors influence sales moderately

Conclusion

This project demonstrates how data analytics and visualization tools can be integrated to build a dynamic sales analysis and forecasting system that supports business decision-making.